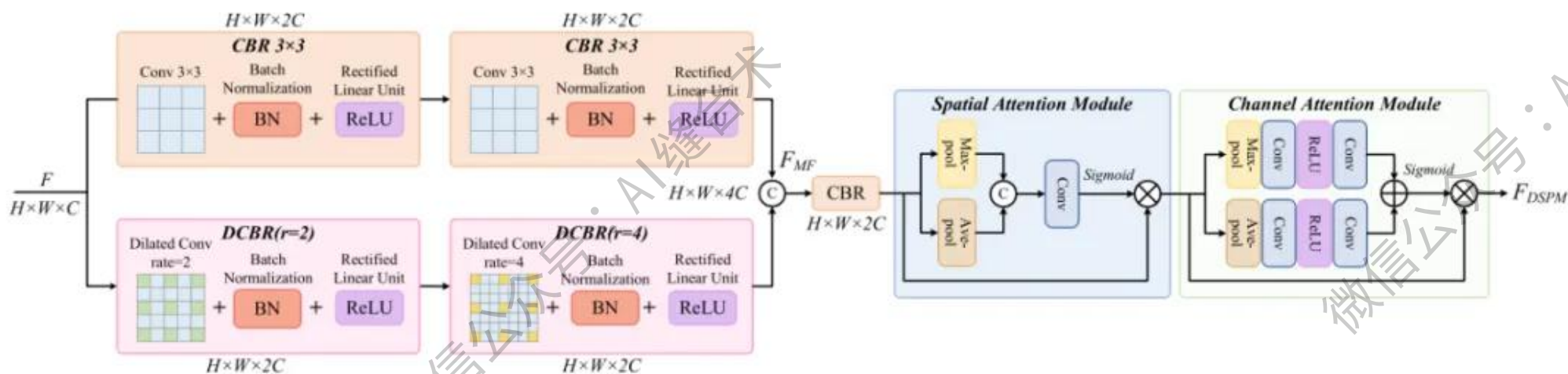


核心速览



论文题目：HAFNet: Hierarchical Attention Fusion Network for Infrared Small Target Detection

中文题目：HAFNet：红外小目标检测的层次注意力融合网络

论文链接：

<https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=11154006>

所属单位：江西财经大学等

核心速览：本文提出了一种基于 U-Net 架构的新型分层注意力融合网络（HAFNet），用于红外小目标检测（IRSTD），通过设计双分支语义感知模块和分层特征融合编解码器，显著提升了复杂场景中小目标的检测性能。

论文内容详细解读：<https://mp.weixin.qq.com/s/xTrujJX86hYeT1DsKud6A>

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```
,
(dconv_layer): Sequential(
  (0): Sequential(
    (0): Conv2d(32, 32, kernel_size=(3, 3), stride=(1, 1), padding=(2, 2), dilation=(2, 2))
    (1): BatchNorm2d(32, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
    (2): ReLU(inplace=True)
  )
  (1): Sequential(
    (0): Conv2d(32, 32, kernel_size=(3, 3), stride=(1, 1), padding=(4, 4), dilation=(4, 4))
    (1): BatchNorm2d(32, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
    (2): ReLU(inplace=True)
  )
)
(final_layer): Sequential(
  (0): Conv2d(64, 32, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1))
  (1): BatchNorm2d(32, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
  (2): ReLU(inplace=True)
)
(ca): ChannelAttention(
  (avg_pool): AdaptiveAvgPool2d(output_size=1)
  (max_pool): AdaptiveMaxPool2d(output_size=1)
  (fc1): Conv2d(32, 2, kernel_size=(1, 1), stride=(1, 1), bias=False)
  (relu1): ReLU()
  (fc2): Conv2d(2, 32, kernel_size=(1, 1), stride=(1, 1), bias=False)
  (sigmoid): Sigmoid()
)
(sa): SpatialAttention(
  (conv1): Conv2d(2, 1, kernel_size=(7, 7), stride=(1, 1), padding=(3, 3), bias=False)
  (sigmoid): Sigmoid()
)
)
```

哔哩哔哩/微信公众号：AI缝合术，独家整理！

Input : torch.Size([1, 32, 128, 128])
Output: torch.Size([1, 32, 128, 128])

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